

Introduction to Artificial Intelligence and RAG

1. Artificial Intelligence

Artificial Intelligence (AI) is a field of computer science that focuses on creating systems that can perform tasks that normally require human intelligence. These tasks include learning, reasoning, understanding language, recognizing patterns, and making decisions.

2. Machine Learning

Machine Learning (ML) is a subset of Artificial Intelligence. Instead of explicitly programming every rule, a machine learning system learns patterns from data. Common machine learning tasks include classification, regression, clustering, and prediction.

3. Generative AI

Generative AI refers to AI systems that can create new content such as text, images, audio, or code. Large Language Models (LLMs) are a type of generative AI system designed to understand and generate natural language.

4. Retrieval-Augmented Generation

Retrieval-Augmented Generation (RAG) combines information retrieval with text generation. A typical RAG system first retrieves relevant information from a knowledge source and then provides that information to a language model as context. The language model uses the retrieved context to generate an answer.

5. RAG Pipeline

A basic RAG pipeline contains several steps. First, a document is loaded and its text is extracted. The text is divided into smaller chunks. Each chunk is converted into a numerical representation called an embedding. The embeddings are stored in a vector index. When a user asks a question, the question is also converted into an embedding. The system searches the vector index and retrieves the most relevant chunks. These chunks are then supplied to the language model as context.

6. Benefits of RAG

RAG can help a language model answer questions using information from a specific document or knowledge base. It can make answers more grounded in the supplied information and can provide source references such as document pages. RAG can also be useful when the information is specific to an organization or document collection.

7. Limitations of RAG

A RAG system depends on the quality of document extraction, chunking, embeddings, and retrieval. If the relevant information is not retrieved, the language model may not have enough context to answer correctly. Scanned documents may also require OCR before their text can be retrieved.

8. Example

Suppose a company stores its employee handbook as a PDF. An employee asks, 'How many days of annual leave are provided?' A RAG system can retrieve the section of the handbook containing the leave policy and provide that section to the language model. The model can then generate a concise answer based on the retrieved text.